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Java SE 27 & JDK 27

Class SelectionKey

java.lang.Object
    java.nio.channels.SelectionKey

Direct Known Subclasses:

AbstractSelectionKey

```
public abstract class SelectionKey
extends Object
```

A token representing the registration of a `SelectableChannel` with a `Selector`.

A selection key is created each time a channel is registered with a selector. A key remains valid until it is *cancelled* by invoking its `cancel` method, by closing its channel, or by closing its selector. Cancelling a key does not immediately remove it from its selector; it is instead added to the selector's *cancelled-key set* for removal during the next selection operation. The validity of a key may be tested by invoking its `isValid` method.

A selection key contains two *operation sets* represented as integer values. Each bit of an operation set denotes a category of selectable operations that are supported by the key's channel.

- The *interest set* determines which operation categories will be tested for readiness the next time one of the selector's selection methods is invoked. The interest set is initialized with the value given when the key is created; it may later be changed via the `interestOps(int)` method.
- The *ready set* identifies the operation categories for which the key's channel has been detected to be ready by the key's selector. The ready set is initialized to zero when the key is created; it may later be updated by the selector during a selection operation, but it cannot be updated directly.

That a selection key's ready set indicates that its channel is ready for some operation category is a hint, but not a guarantee, that an operation in such a category may be performed by a thread without causing the thread to block. A ready set is most likely to be accurate immediately after the completion of a selection operation. It is likely to be made inaccurate by external events and by I/O operations that are invoked upon the corresponding channel.

This class defines all known operation-set bits, but precisely which bits are supported by a given channel depends upon the type of the channel. Each subclass of `SelectableChannel` defines a `validOps()` method which returns a set identifying just those operations that are supported by the channel. An attempt to set or test an operation-set bit that is not supported by a key's channel will result in an appropriate run-time exception.

It is often necessary to associate some application-specific data with a selection key, for example an object that represents the state of a higher-level protocol and handles readiness notifications in order to implement that protocol. Selection keys therefore support the *attachment* of a single arbitrary object to a key. An object can be attached via the `attach` method and then later retrieved via the `attachment` method.

Selection keys are safe for use by multiple concurrent threads. A selection operation will always use the interest-set value that was current at the moment that the operation began.

Since:

1.4

See Also:

`SelectableChannel`, `Selector`

Field Summary

Fields

Modifier and Type	Field	Description
`static final int`	**`OP_ACCEPT`**	Operation-set bit for socket-accept operations.
`static final int`	**`OP_CONNECT`**	Operation-set bit for socket-connect operations.
`static final int`	**`OP_READ`**	Operation-set bit for read operations.
`static final int`	**`OP_WRITE`**	Operation-set bit for write operations.

Constructor Summary

Constructors

Modifier	Constructor	Description
`protected`	**`SelectionKey()`**	Constructs an instance of this class.

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
final Object	**attach**(Object ob)	Attaches the given object to this key.
final Object	**attachment**()	Retrieves the current attachment.
abstract void	**cancel**()	Requests that the registration of this key's channel with its selector be cancelled.
abstract SelectableChannel	**channel**()	Returns the channel for which this key was created.
abstract int	**interestOps**()	Retrieves this key's interest set.
abstract SelectionKey	**interestOps**(int ops)	Sets this key's interest set to the given value.
int	**interestOpsAnd**(int ops)	Atomically sets this key's interest set to the bitwise intersection ("and") of the existing interest set and the given value.
int	**interestOpsOr**(int ops)	Atomically sets this key's interest set to the bitwise union ("or") of the existing interest set and the given value.
final boolean	**isAcceptable**()	Tests whether this key's channel is ready to accept a new socket connection.
final boolean	**isConnectable**()	Tests whether this key's channel has either finished, or failed to

		finish, its socket-connection operation.
final boolean	**isReadable**()	Tests whether this key's channel is ready for reading.
abstract boolean	**isValid**()	Tells whether or not this key is valid.
final boolean	**isWritable**()	Tests whether this key's channel is ready for writing.
abstract int	**readyOps**()	Retrieves this key's ready-operation set.
abstract Selector	**selector**()	Returns the selector for which this key was created.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Field Details

OP_READ

```
public static final int OP_READ
```

Operation-set bit for read operations.

Suppose that a selection key's interest set contains `OP_READ` at the start of a selection operation. If the selector detects that the corresponding channel is ready for reading, has reached end-of-stream, has been remotely shut down for further writing, or has an error pending, then it will add `OP_READ` to the key's ready-operation set.

See Also:

Constant Field Values

OP_WRITE

```
public static final int OP_WRITE
```

Operation-set bit for write operations.

Suppose that a selection key's interest set contains `OP_WRITE` at the start of a selection operation. If the selector detects that the corresponding channel is ready for writing, has been remotely shut down for further reading, or has an error pending, then it will add `OP_WRITE` to the key's ready set.

See Also:

Constant Field Values

OP_CONNECT

```
public static final int OP_CONNECT
```

Operation-set bit for socket-connect operations.

Suppose that a selection key's interest set contains `OP_CONNECT` at the start of a selection operation. If the selector detects that the corresponding socket channel is ready to complete its connection sequence, or has an error pending, then it will add `OP_CONNECT` to the key's ready set.

See Also:

Constant Field Values

OP_ACCEPT

```
public static final int OP_ACCEPT
```

Operation-set bit for socket-accept operations.

Suppose that a selection key's interest set contains `OP_ACCEPT` at the start of a selection operation. If the selector detects that the corresponding server-socket channel is ready to accept another connection, or has an error pending, then it will add `OP_ACCEPT` to the key's ready set.

See Also:

Constant Field Values

Constructor Details

SelectionKey

```
protected SelectionKey()
```

Constructs an instance of this class.

Method Details

channel

```
public abstract SelectableChannel channel()
```

Returns the channel for which this key was created. This method will continue to return the channel even after the key is cancelled.

Returns:

This key's channel

selector

```
public abstract Selector selector()
```

Returns the selector for which this key was created. This method will continue to return the selector even after the key is cancelled.

Returns:

This key's selector

isValid

```
public abstract boolean isValid()
```

Tells whether or not this key is valid.

A key is valid upon creation and remains so until it is cancelled, its channel is closed, or its selector is closed.

Returns:

`true` if, and only if, this key is valid

cancel

```
public abstract void cancel()
```

Requests that the registration of this key's channel with its selector be cancelled. Upon return the key will be invalid and will have been added to its selector's cancelled-key set. The key will be removed from all of the selector's key sets during the next selection operation.

If this key has already been cancelled then invoking this method has no effect. Once cancelled, a key remains forever invalid.

This method may be invoked at any time. It synchronizes on the selector's cancelled-key set, and therefore may block briefly if invoked concurrently with a cancellation or selection operation involving the same selector.

interestOps

```
public abstract int interestOps()
```

Retrieves this key's interest set.

It is guaranteed that the returned set will only contain operation bits that are valid for this key's channel.

Returns:

This key's interest set

Throws:

`CancelledKeyException` - If this key has been cancelled

interestOps

```
public abstract SelectionKey interestOps(int ops)
```

Sets this key's interest set to the given value.

This method may be invoked at any time. If this method is invoked while a selection operation is in progress then it has no effect upon that operation; the change to the key's interest set will be seen by the next selection operation.

Parameters:

`ops` - The new interest set

Returns:

This selection key

Throws:

`IllegalArgumentException` - If a bit in the set does not correspond to an operation that is supported by this key's channel, that is, if `(ops & ~channel().validOps()) != 0`

`CancelledKeyException` - If this key has been cancelled

interestOpsOr

```
public int interestOpsOr(int ops)
```

Atomically sets this key's interest set to the bitwise union ("or") of the existing interest set and the given value. This method is guaranteed to be atomic with respect to other concurrent calls to this method or to `interestOpsAnd(int)`.

This method may be invoked at any time. If this method is invoked while a selection operation is in progress then it has no effect upon that operation; the change to the key's interest set will be seen by the next selection operation.

Implementation Requirements:

The default implementation synchronizes on this key and invokes `interestOps()` and `interestOps(int)` to retrieve and set this key's interest set.

Parameters:

`ops` - The interest set to apply

Returns:

The previous interest set

Throws:

`IllegalArgumentException` - If a bit in the set does not correspond to an operation that is supported by this key's channel, that is, if `(ops & ~channel().validOps()) != 0`

`CancelledKeyException` - If this key has been cancelled

Since:

11

interestOpsAnd

```
public int interestOpsAnd(int ops)
```

Atomically sets this key's interest set to the bitwise intersection ("and") of the existing interest set and the given value. This method is guaranteed to be atomic with respect to other concurrent calls to this method or to `interestOpsOr(int)`.

This method may be invoked at any time. If this method is invoked while a selection operation is in progress then it has no effect upon that operation; the change to the key's interest set will be seen by the next selection operation.

API Note:

Unlike the `interestOps(int)` and `interestOpsOr(int)` methods, this method does not throw `IllegalArgumentException` when invoked with bits in the interest set that do not correspond to an operation that is supported by this key's channel. This is to allow operation bits in the interest set to be cleared using bitwise complement values, e.g., `interestOpsAnd(~SelectionKey.OP_READ)` will remove the `OP_READ` from the interest set without affecting other bits.

Implementation Requirements:

The default implementation synchronizes on this key and invokes `interestOps()` and `interestOps(int)` to retrieve and set this key's interest set.

Parameters:

`ops` - The interest set to apply

Returns:

The previous interest set

Throws:

`CancelledKeyException` - If this key has been cancelled

Since:

11

readyOps

```
public abstract int readyOps()
```

Retrieves this key's ready-operation set.

It is guaranteed that the returned set will only contain operation bits that are valid for this key's channel.

Returns:

This key's ready-operation set

Throws:

`CancelledKeyException` - If this key has been cancelled

isReadable

```
public final boolean isReadable()
```

Tests whether this key's channel is ready for reading.

An invocation of this method of the form `k.isReadable()` behaves in exactly the same way as the expression

```
    k.readyOps() & OP_READ != 0
```

If this key's channel does not support read operations then this method always returns `false`.

Returns:

`true` if, and only if, `readyOps() & OP_READ` is nonzero

Throws:

CancelledKeyException - If this key has been cancelled

isWritable

```
public final boolean isWritable()
```

Tests whether this key's channel is ready for writing.

An invocation of this method of the form `k.isWritable()` behaves in exactly the same way as the expression

```
    k.readyOps() & OP_WRITE != 0
```

If this key's channel does not support write operations then this method always returns `false`.

Returns:

`true` if, and only if, `readyOps() & OP_WRITE` is nonzero

Throws:

CancelledKeyException - If this key has been cancelled

isConnectable

```
public final boolean isConnectable()
```

Tests whether this key's channel has either finished, or failed to finish, its socket-connection operation.

An invocation of this method of the form `k.isConnectable()` behaves in exactly the same way as the expression

```
    k.readyOps() & OP_CONNECT != 0
```

If this key's channel does not support socket-connect operations then this method always returns `false`.

Returns:

`true` if, and only if, `readyOps() & OP_CONNECT` is nonzero

Throws:

CancelledKeyException - If this key has been cancelled

isAcceptable

```
public final boolean isAcceptable()
```

Tests whether this key's channel is ready to accept a new socket connection.

An invocation of this method of the form `k.isAcceptable()` behaves in exactly the same way as the expression

```
k.readyOps() & OP_ACCEPT != 0
```

If this key's channel does not support socket-accept operations then this method always returns `false`.

Returns:

`true` if, and only if, `readyOps() & OP_ACCEPT` is nonzero

Throws:

`CancelledKeyException` - If this key has been cancelled

attach

`public final Object attach(Object ob)`

Attaches the given object to this key.

An attached object may later be retrieved via the `attachment` method. Only one object may be attached at a time; invoking this method causes any previous attachment to be discarded. The current attachment may be discarded by attaching `null`.

Parameters:

`ob` - The object to be attached; may be `null`

Returns:

The previously-attached object, if any, otherwise `null`

attachment

`public final Object attachment()`

Retrieves the current attachment.

Returns:

The object currently attached to this key, or `null` if there is no attachment

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Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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